University of Basel
Continuing Education

CONTINUING EDUCATION

Chapter 13 – Probabilistic Reasoning

Machine Intelligence Seminar

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Outline

1. **Bayesian Networks:** Introduction & Semantics
2. **Exact Inference:** Introduction & Example
3. **Approximate Inference**
4. **Discussion**

Bayesian Networks

Why Probabilistic Reasoning?

- Real-world environments are **uncertain**.

Uncertain environments:

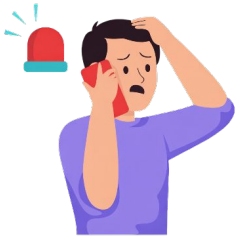
- Incomplete information
- Noisy observations
- Unpredictable events

Example: Burglary Alarm

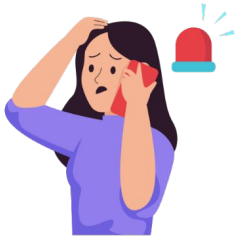
Burglary

Earthquake

Alarm

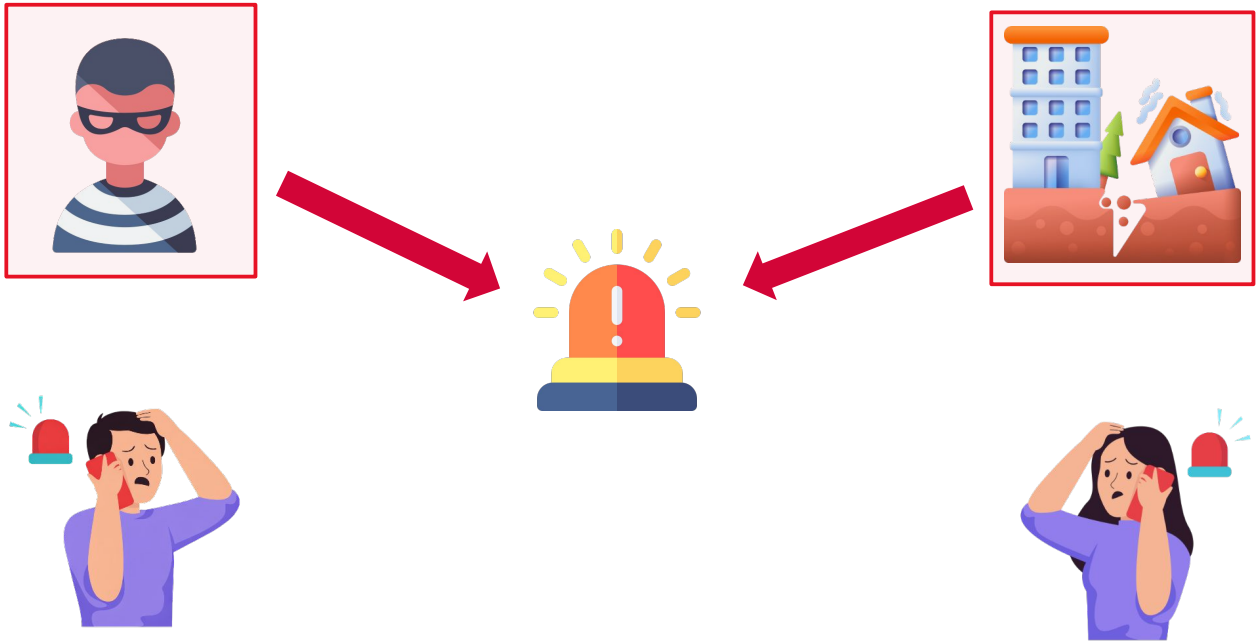


John



Mary

Example: Burglary Alarm



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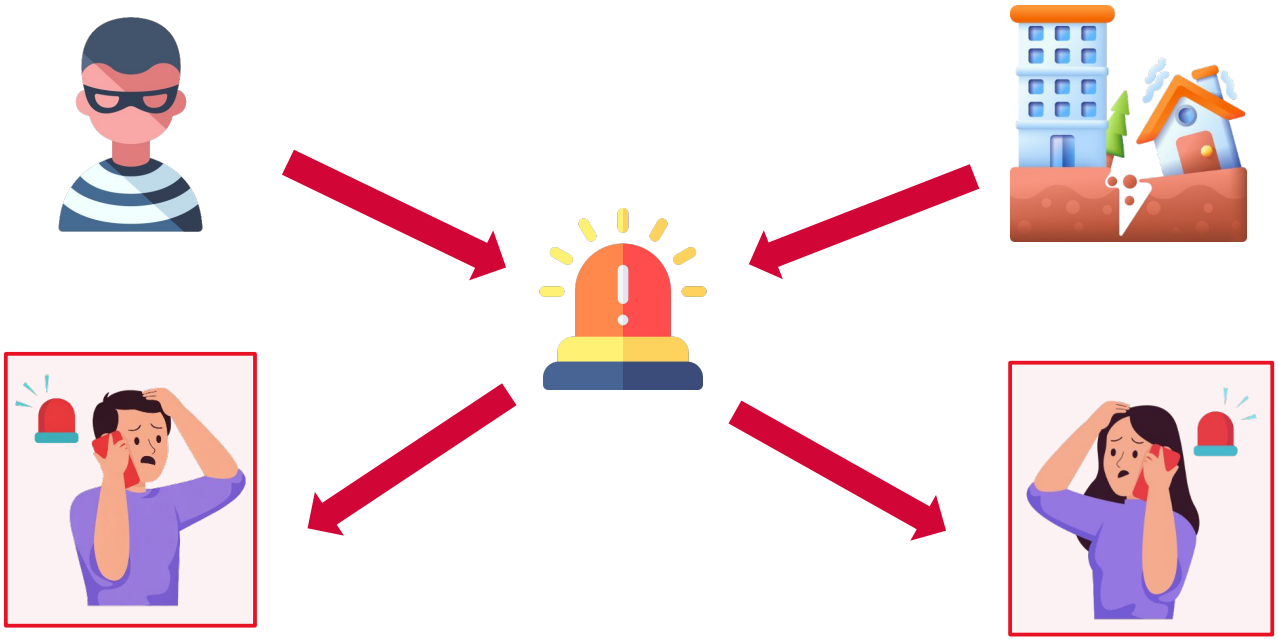


Full joint probability distribution

- Full joint probability distribution: $P(X_1, X_2, ..., X_n)$
- *n* boolean variables → 2^n probabilities
- In our case n=5 ⇒ **32 probabilities**

↳ **Problem:** *Number of probabilities grows exponentially!*

- n = 30 → 2^{30} ≈ 1,000,000,000 ⇒ **intractable**

↳ We need a more compact representation

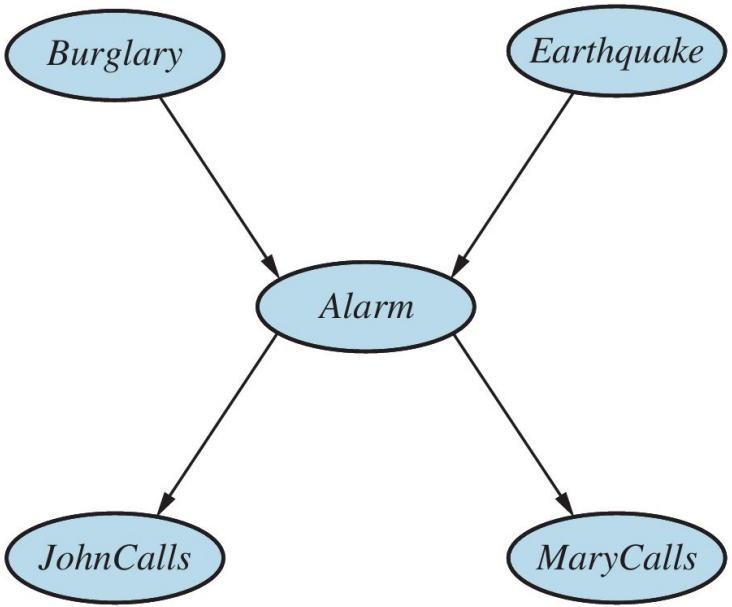
Definition: Bayesian Network

- Instead of listing all probabilities for every possible combination of variables, we only store **local conditional probabilities**

$$P(x_1, ..., x_n) = \prod_{i=1}^{n} P(x_i \mid Parents(X_i))$$

Semantics: Bayesian Network

- Is a **D**irected **A**cyclic **G**raph (DAG)



- **Nodes** = random variables
- **Edges** = direct probabilistic influence
- **Conditional probability table** for each node

Semantics: Bayesian Network

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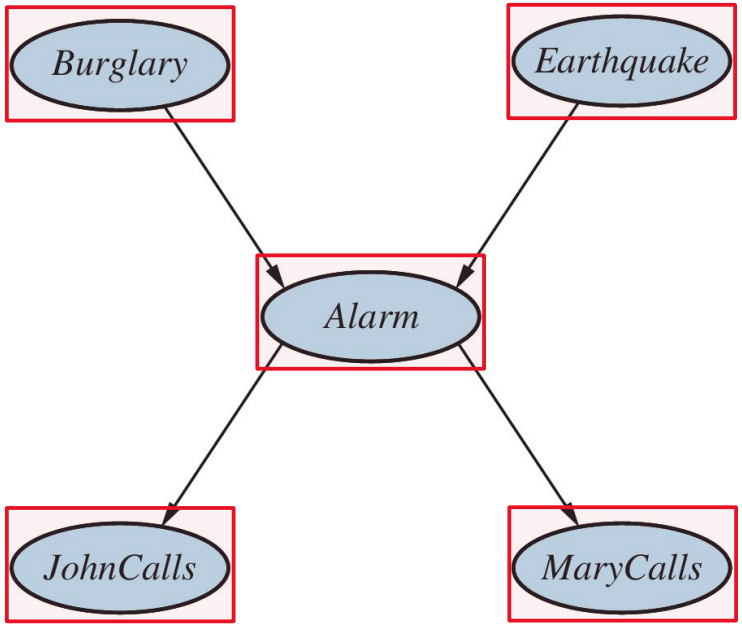

- **Nodes** = {*Burglary, Earthquake, Alarm, JohnCalls, MaryCalls*}
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Semantics: Bayesian Network

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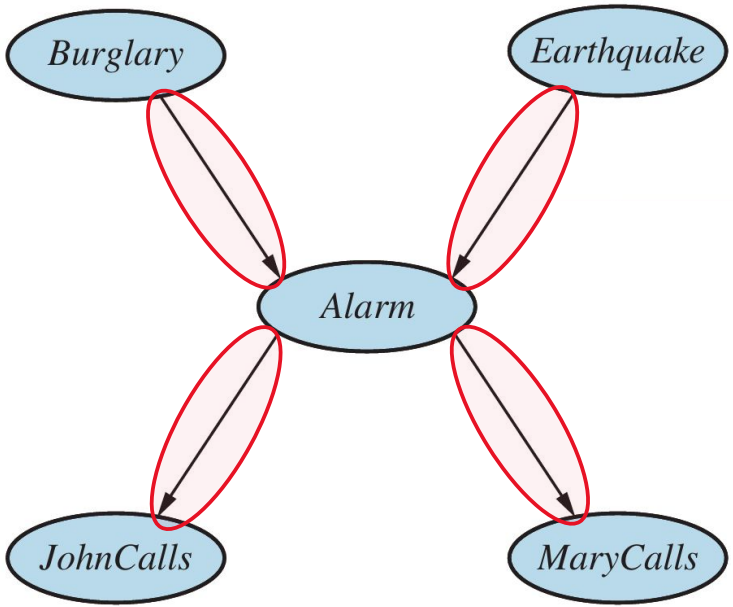

- **Nodes** = {*Burglary, Earthquake, Alarm, JohnCalls, MaryCalls*}
- **Edges** = {*Burglary* → *Alarm, Earthquake* → *Alarm, Alarm* → *JohnCalls, Alarm* → *MaryCalls*}
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Semantics: Bayesian Network

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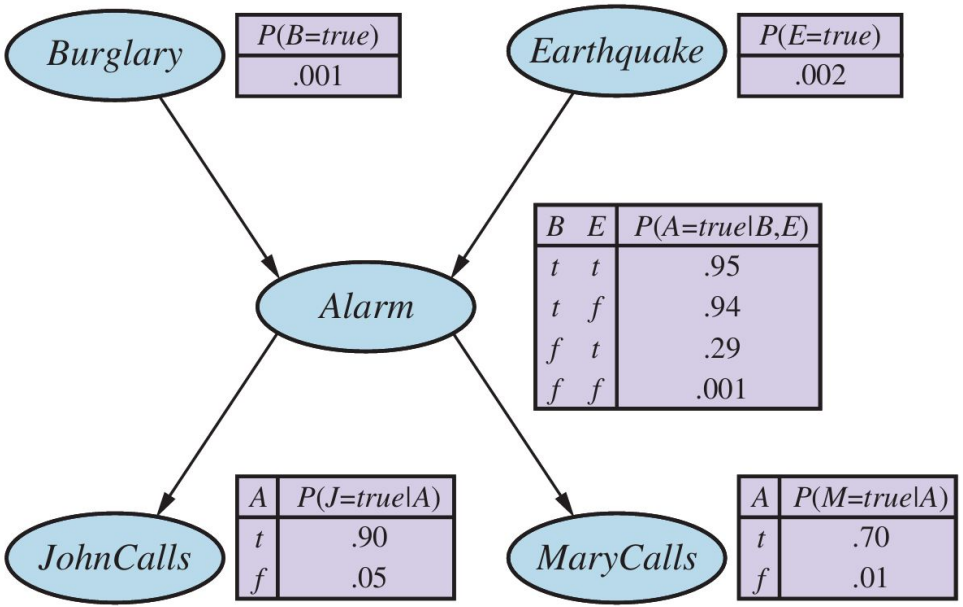

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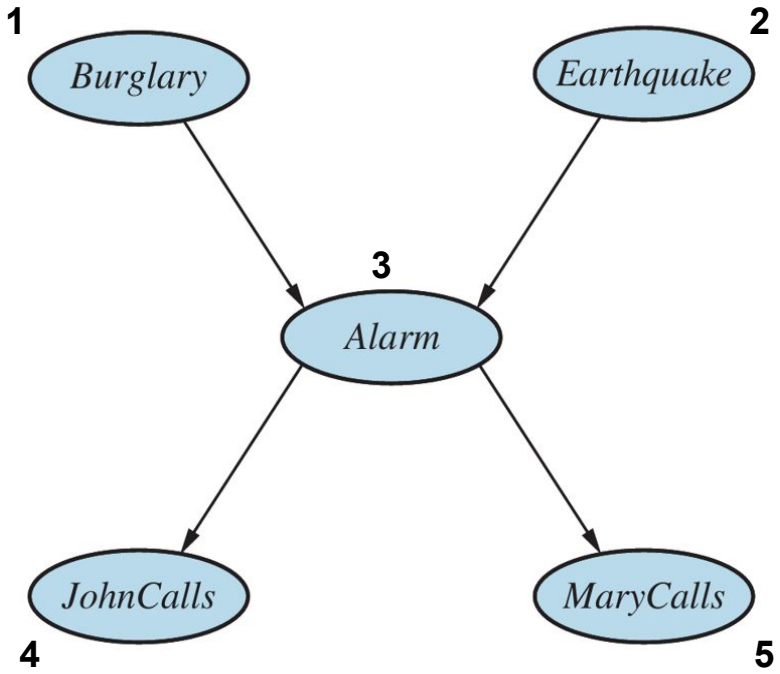
- **Edges** = {*Burglary* → *Alarm, Earthquake* → *Alarm, Alarm* → *JohnCalls, Alarm* → *MaryCalls*}
- **Conditional probability table for each node**

Constructing the graph: Node order

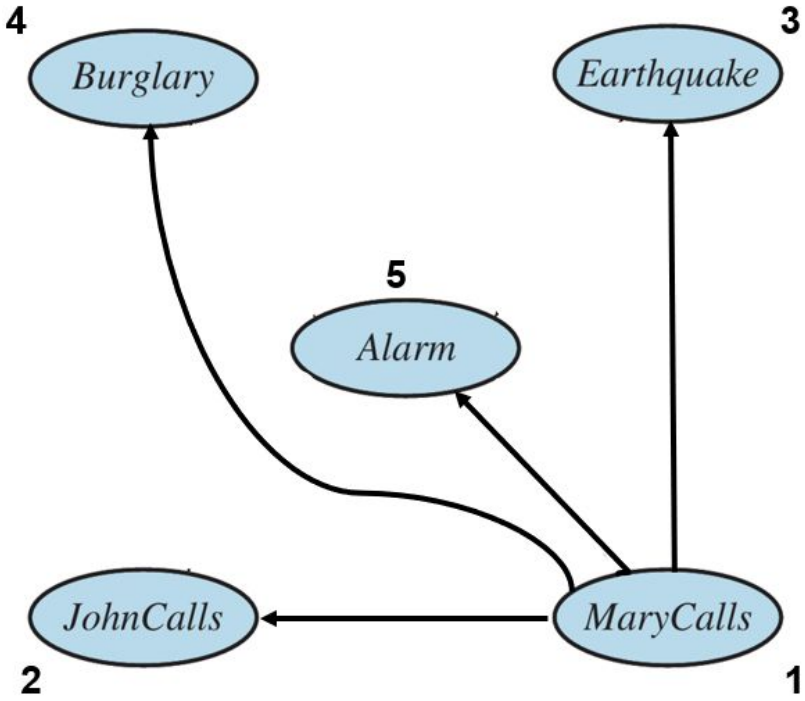
- Compactness depends on node order
- **Parents** are chosen from earlier variables in the order → **Causal Order**



Requires 10 parameters

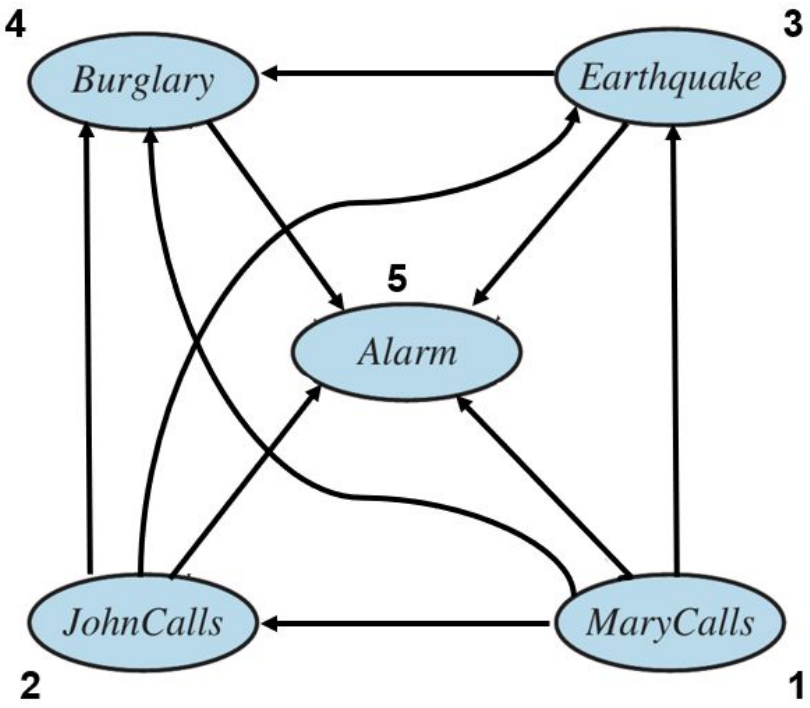
What if the order is different?

- Different variable orderings → Different network structures



Bad Order

- Loss of independence → Variables now depend on many more parents



Requires 31 parameters

Formalizing the Bayesian Network

$$P(x_1, x_2, \ldots, x_n)$$

Chain Rule

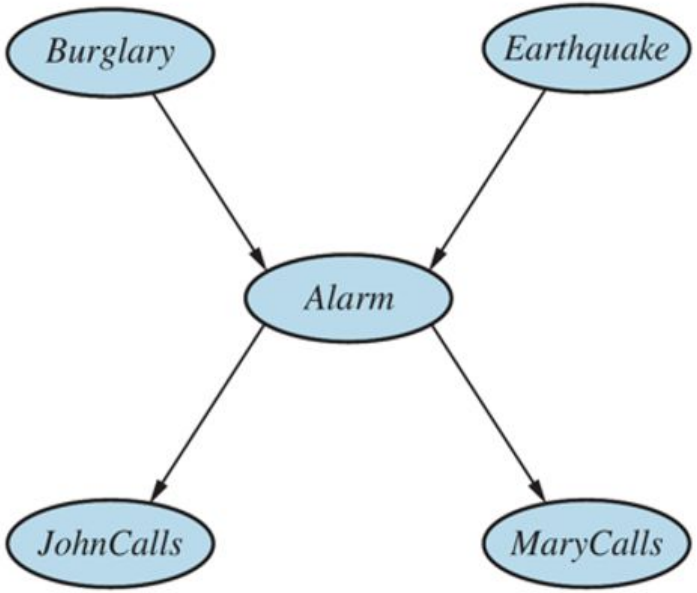
$$= \prod_{i=1}^{n} P(x_i \mid x_{i-1}, \ldots, x_1)$$

Conditional Independence

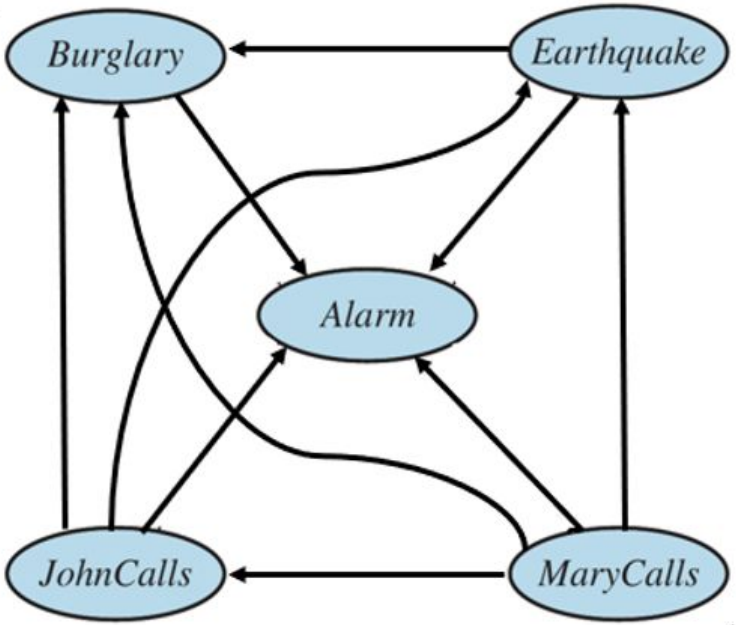
$$= \prod_{i=1}^{n} P(x_i \mid Parents(X_i))$$

Same Distribution, Different Structures

- Joint distribution can be factorized in different valid ways



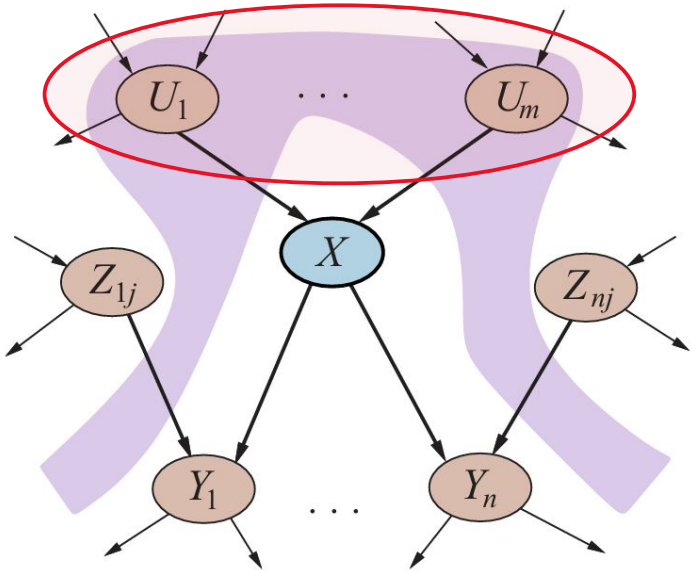
(a)



(b)

Local Conditional Independence

- A variable is independent of its non-descendants given its parents



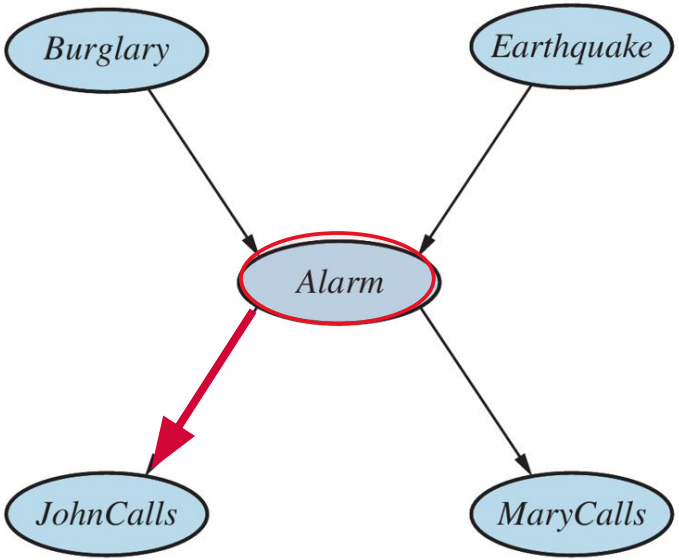
$$X \perp \text{NonDescendants}(X) \mid Parents(X)$$

- Only the parents are needed to determine $P(X)$

Efficient Reasoning

Local Conditional Independence of “JohnCalls”

- A variable is independent of its non-descendants given its parents



- JohnCalls depends only on **Alarm**

$$P(J \mid A, B, E) = P(J \mid A)$$

- **Burglary** and **Earthquake** provide no additional information about JohnCalls

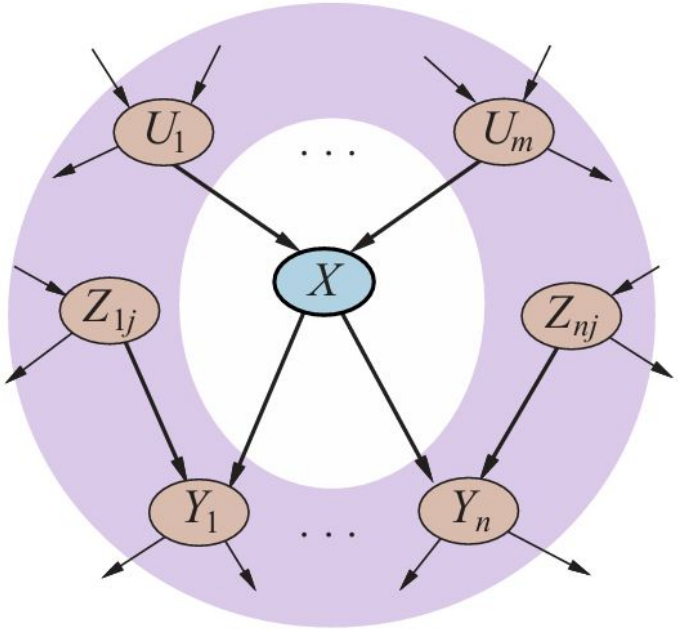
Markov Blanket

- A node X is conditionally independent of all other nodes in the network given its **Markov blanket**

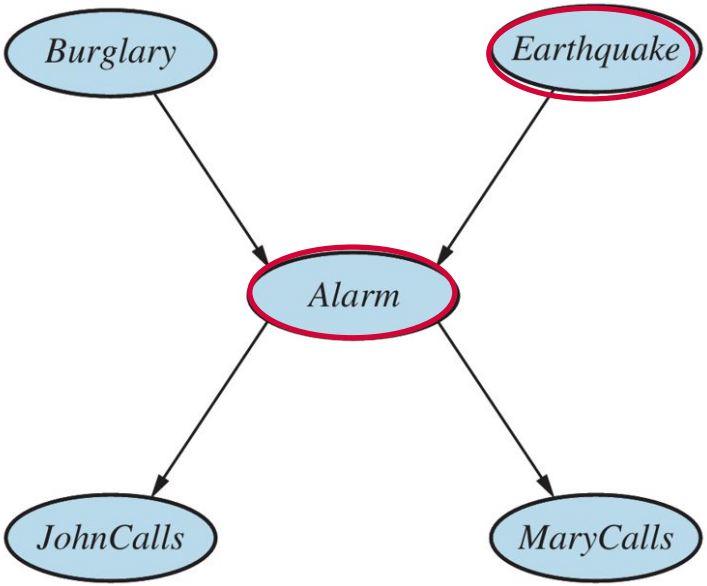
Markov blanket of X:

- Parents of X
- Children of X
- Other parents of the children

All information relevant to X



Markov Blanket of “Burglary”



- **Parents** = none
- **Children** = Alarm
- **Other parent of children** = Earthquake

- *MB(B) = {Alarm, Earthquake}*

Why Bayesian Networks are more Efficient

- **Exploiting Conditional Independence** → Reduces the number of required parameters

- **Full joint distribution** → 2^n probabilities required
- **Bayesian Network** → n · 2^k where k = number of **parents**

  - P(B)
  - P(E)
  - P(A | B,E)
  - P(J | A)
  - P(M | A)

  Only 10 numbers needed

  $n \cdot 2^k < 2^n$

Exact Inference for Bayesian Networks

Exact Inference in Bayesian Networks

```
P(Burglary | JohnCalls = true, MaryCalls = true) =
<0.284, 0.716>
```

- Observe JohnCalls and MaryCalls
- Infer the probability that a burglary occurred with exact numbers

Goal: Compute the probability of some variable when we observe evidence.

Exact Inference in Bayesian Networks

- Inference by Enumeration Algorithm
- The Variable Elimination Algorithm

Inference by Enumeration

Query variable

Evidence Variable

```
P ( X | E )
```

```
P ( Burglary | JohnCalls , MaryCalls )
```

What's the probability that **Burglary** happened, given that **John Called** and **Mary Called**?

Inference by Enumeration

Query variable

Evidence Variable

```
P ( X | E )
```

```
P ( Burglary | JohnCalls , MaryCalls )
```

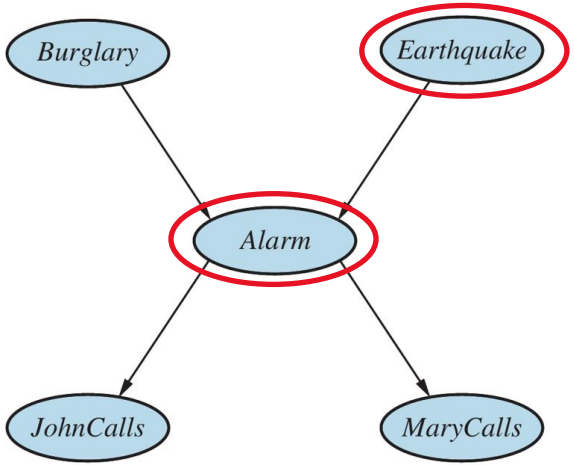
What's the probability that **Burglary** happened, given that **John Called** and **Mary Called**?

Were Burglary, JohnCalls, and MaryCalls the only variables in our example?

Inference by Enumeration

`P ( Burglary | JohnCalls , MaryCalls )`

In our example, Alarm and Earthquake are **hidden variables**



Get probability by summing through all hidden variables from the joint probability distribution

Inference by Enumeration

$$\mathbf{P}(B \mid j,m) = \alpha\, \mathbf{P}(B,j,m) = \alpha \sum_e \sum_a \mathbf{P}(B,j,m,e,a)$$

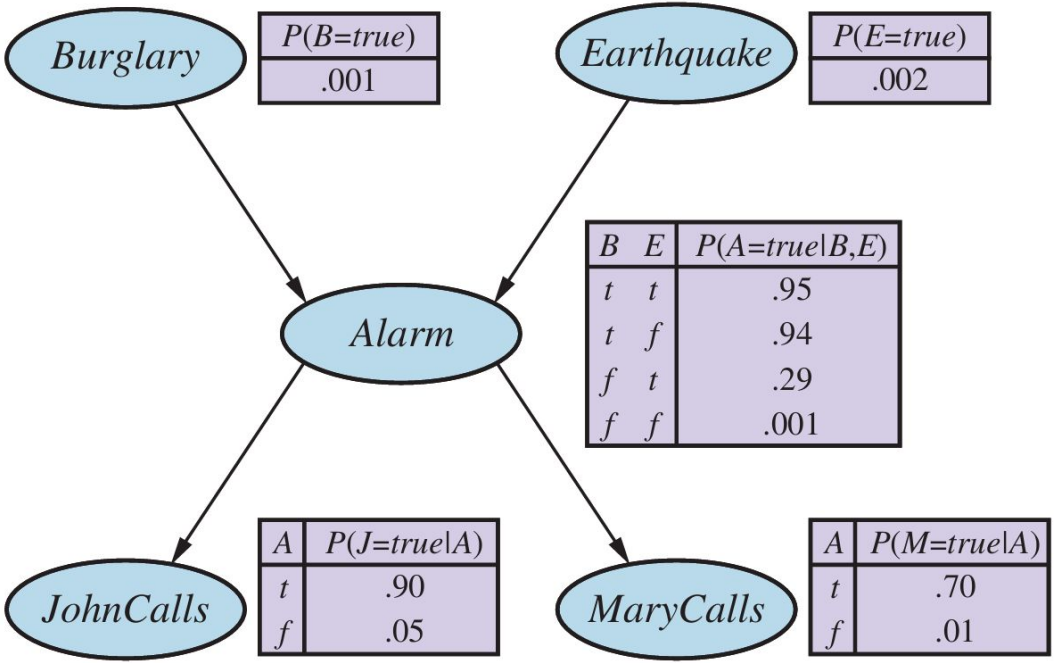
$$\downarrow$$

$$P(b \mid j,m) = \alpha \sum_e \sum_a P(b)P(e)P(a \mid b,e)P(j \mid a)P(m \mid a)$$

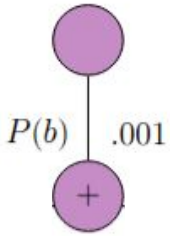
$$\downarrow$$

$$P(b \mid j,m) = \alpha P(b) \sum_e P(e) \sum_a P(a \mid b,e)P(j \mid a)P(m \mid a)$$

Remember Directed Acyclic Graph (DAG)?



Inference by Enumeration Visualization



$$P(b \mid j,m) = \alpha P(b) \sum_{e} P(e) \sum_{a} P(a \mid b,e) P(j \mid a) P(m \mid a)$$

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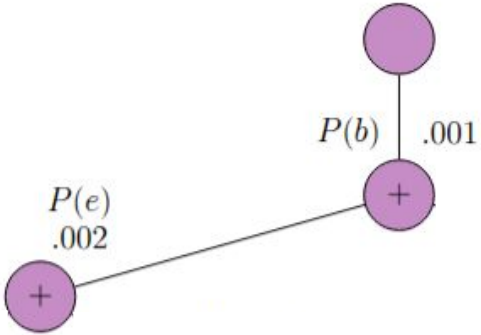

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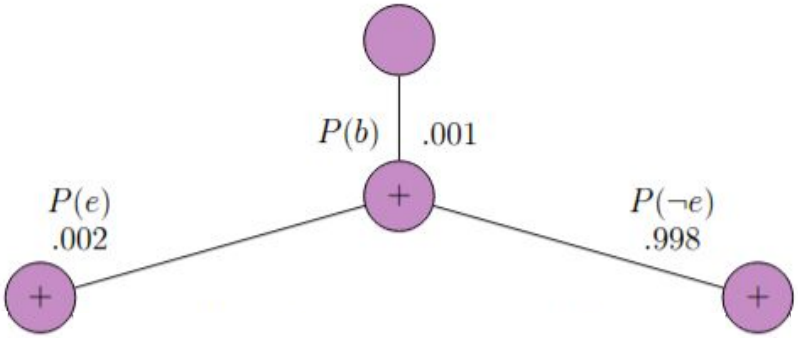

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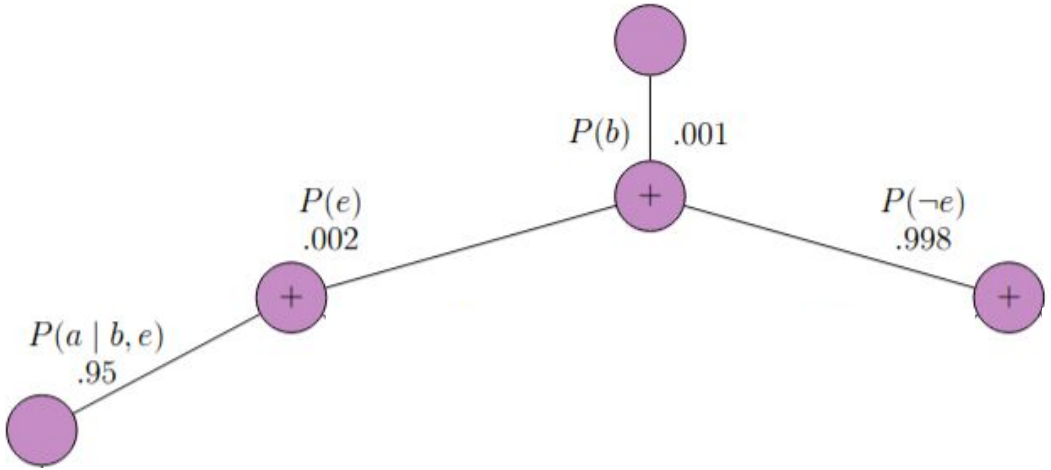

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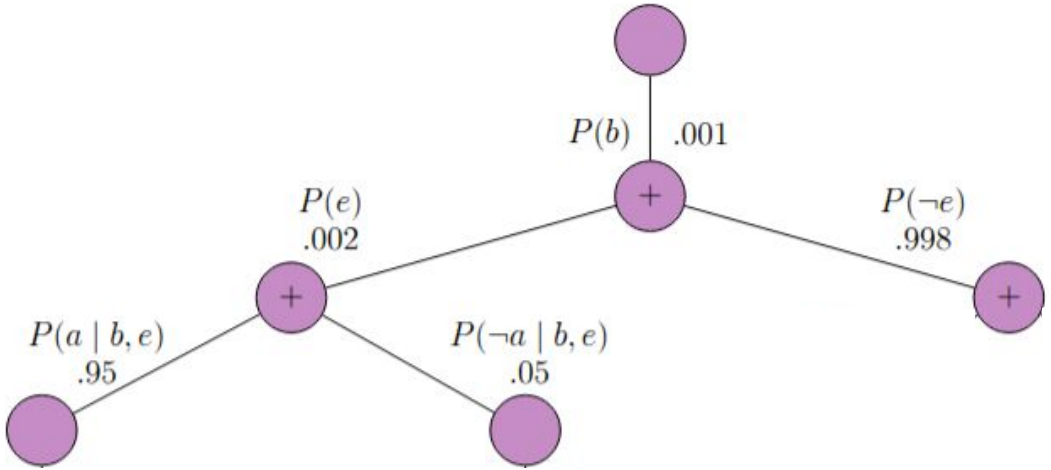

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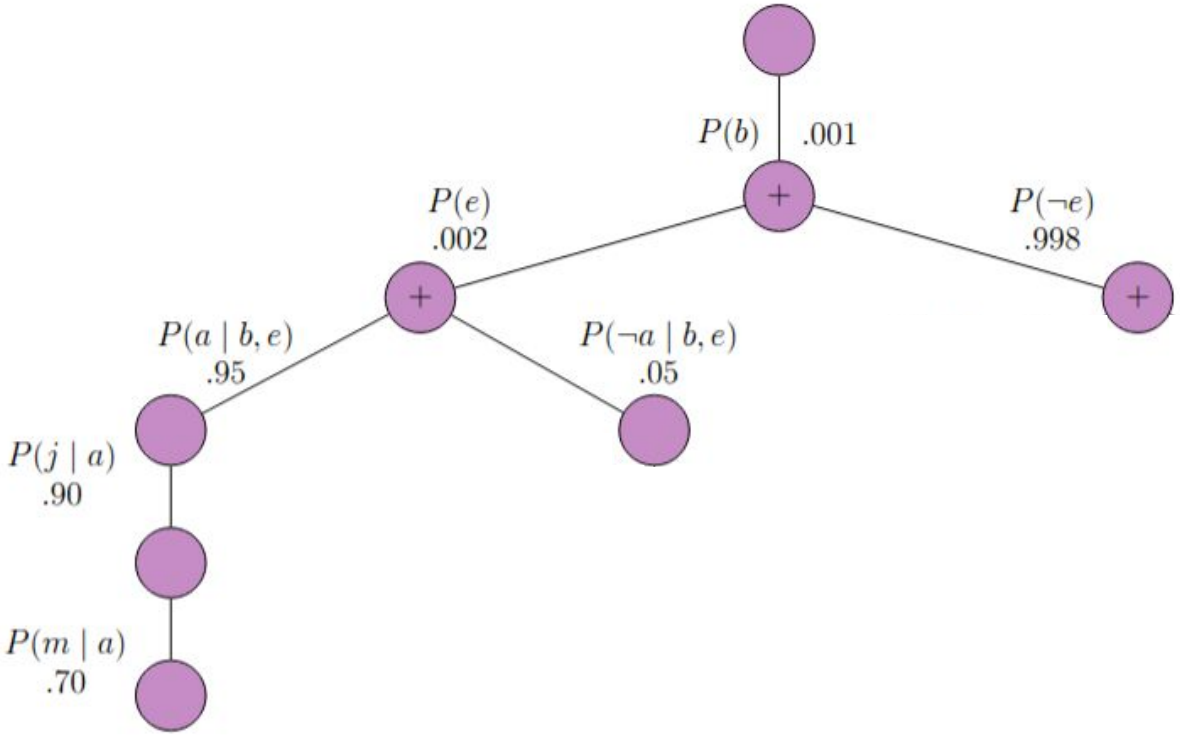

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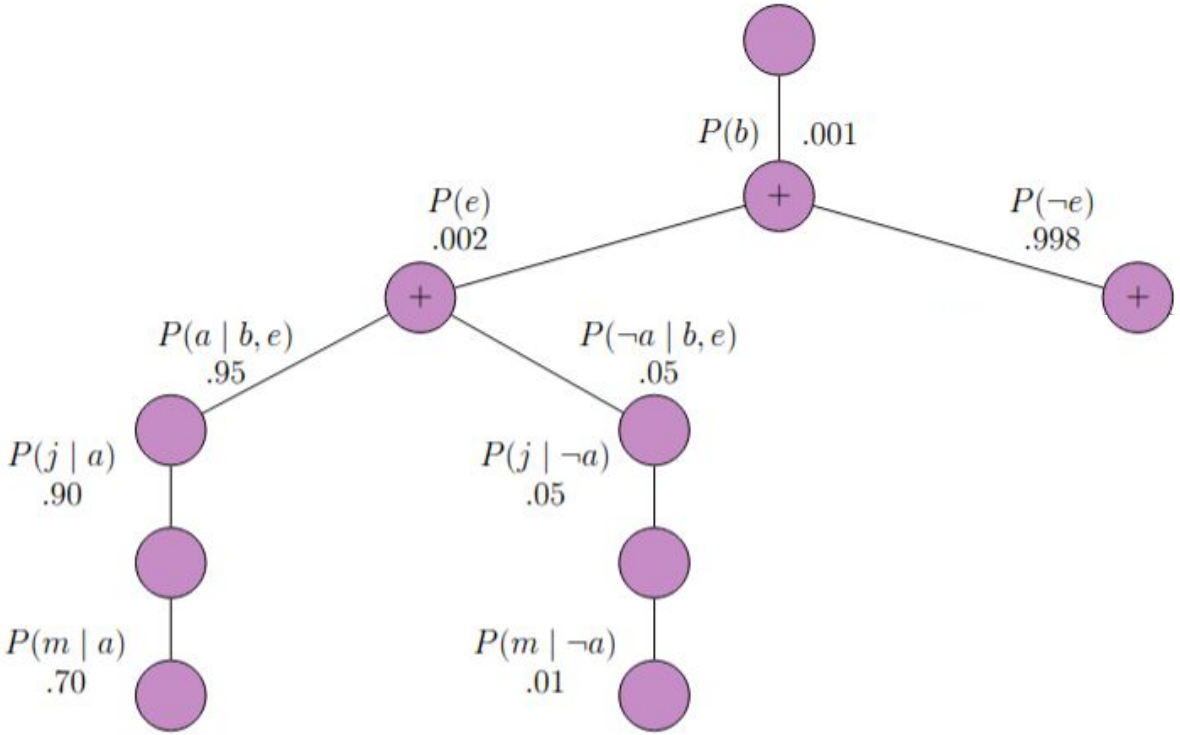

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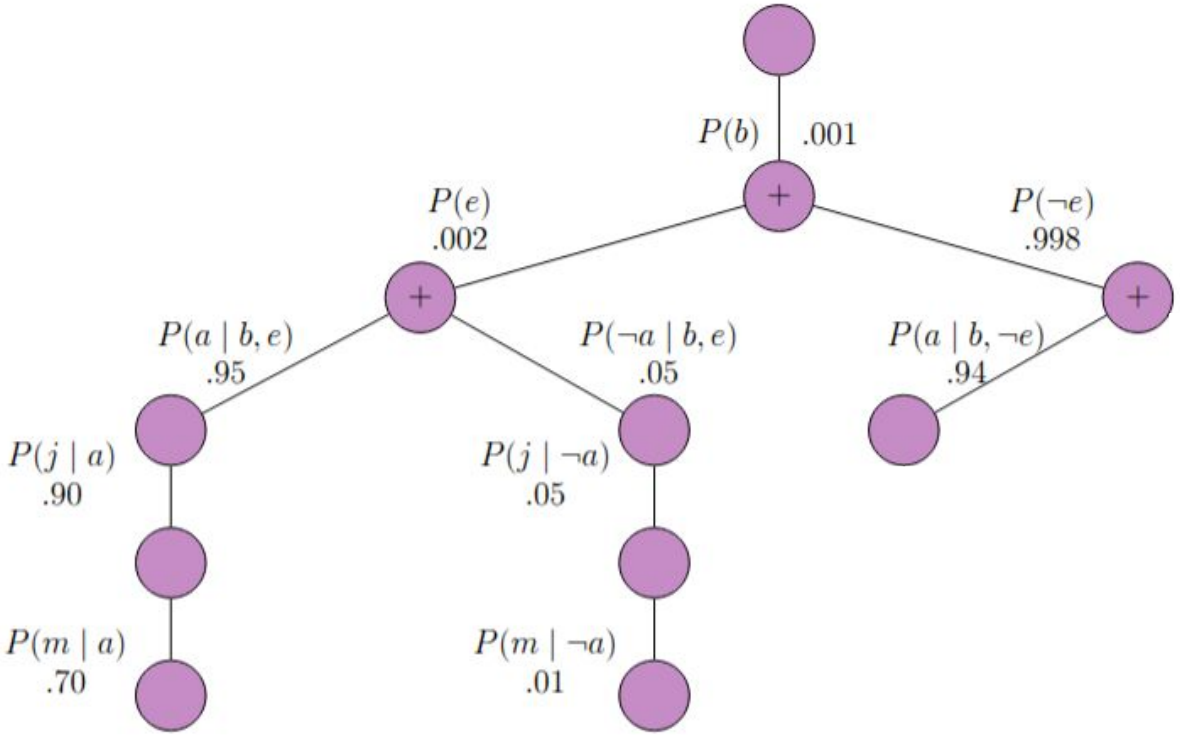

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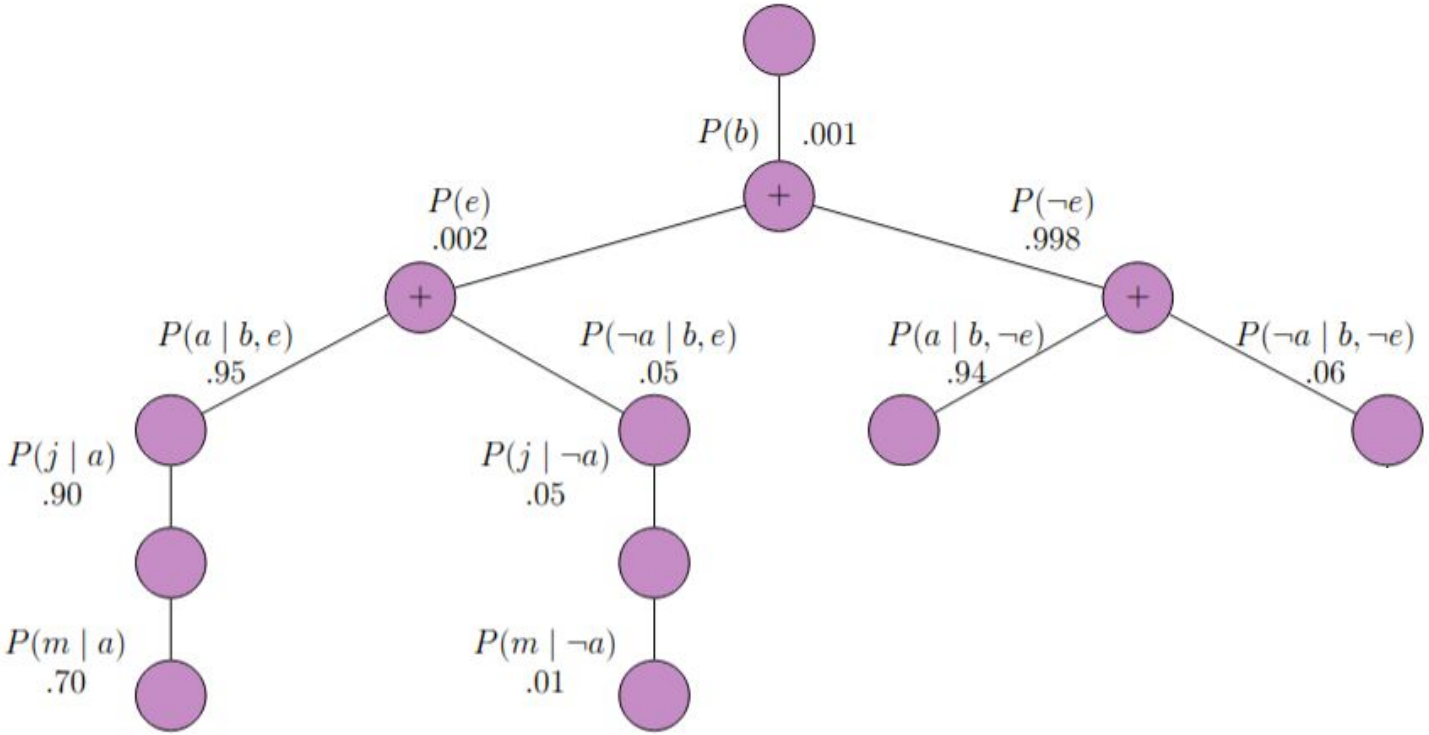

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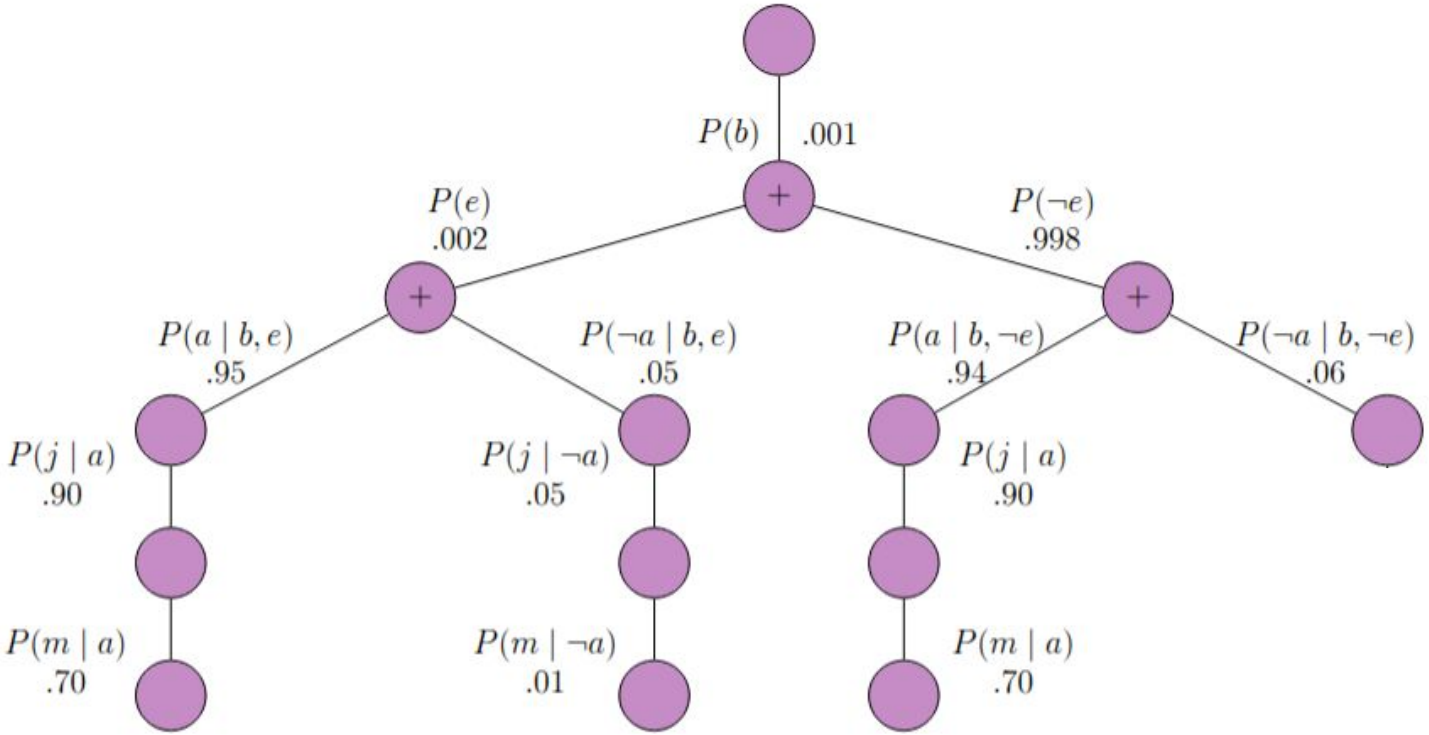

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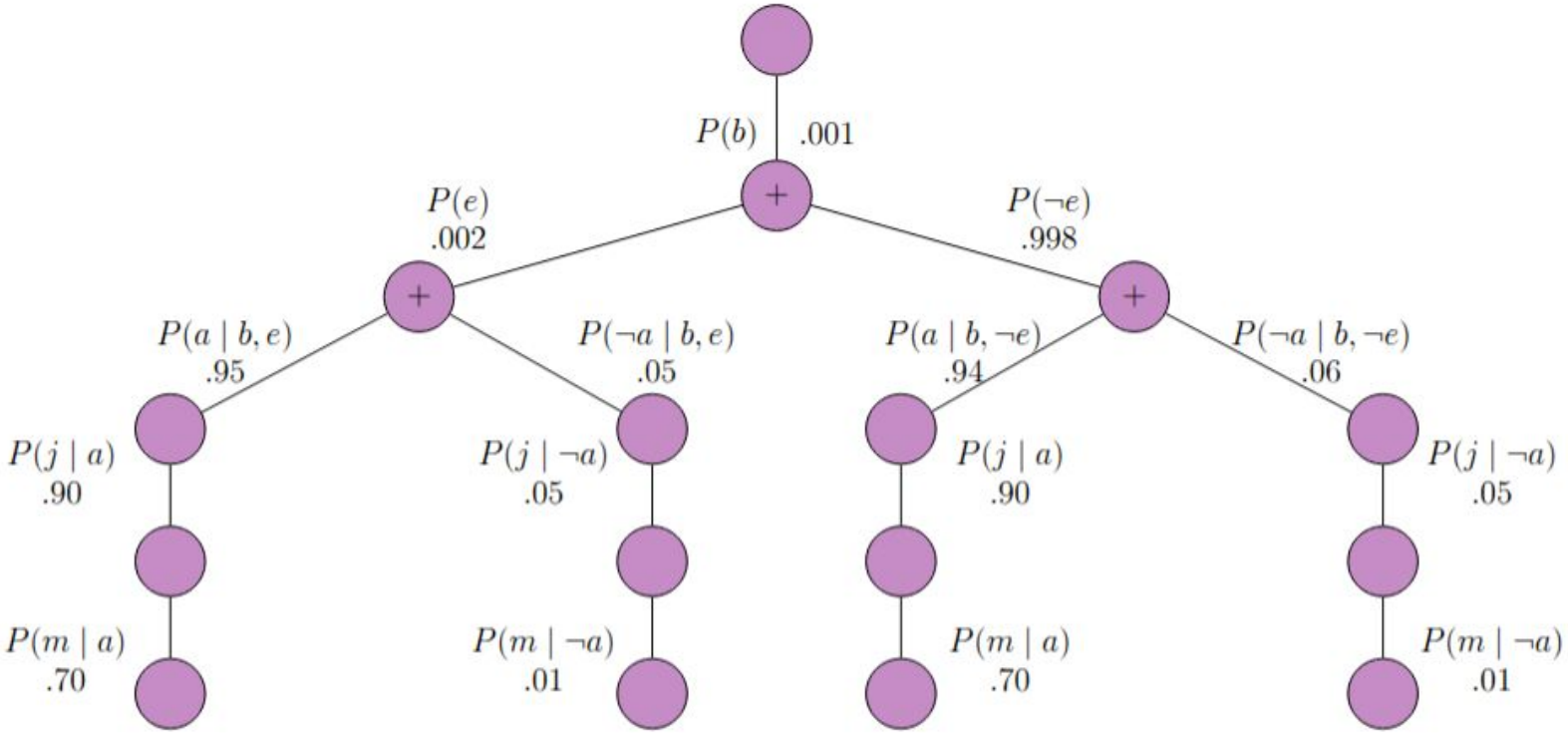

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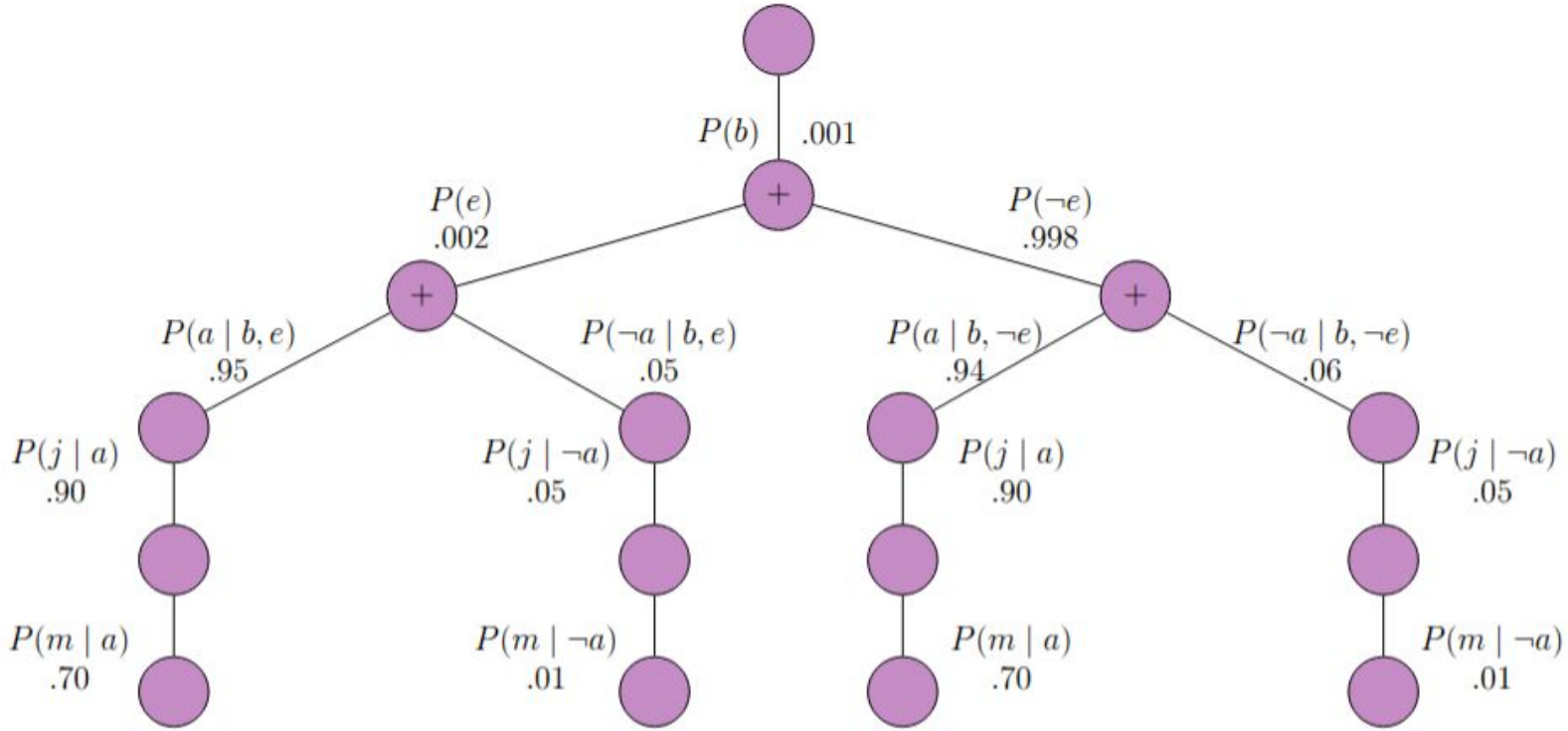


$$\mathbf{P}(Burglary \mid JohnCalls = true, MaryCalls = true) = \langle 0.284, 0.716 \rangle$$

Complexity of Inference by Enumeration

$$P(b \mid j,m) = \alpha P(b) \sum_e P(e) \sum_a P(a \mid b,e) P(j \mid a) P(m \mid a)$$

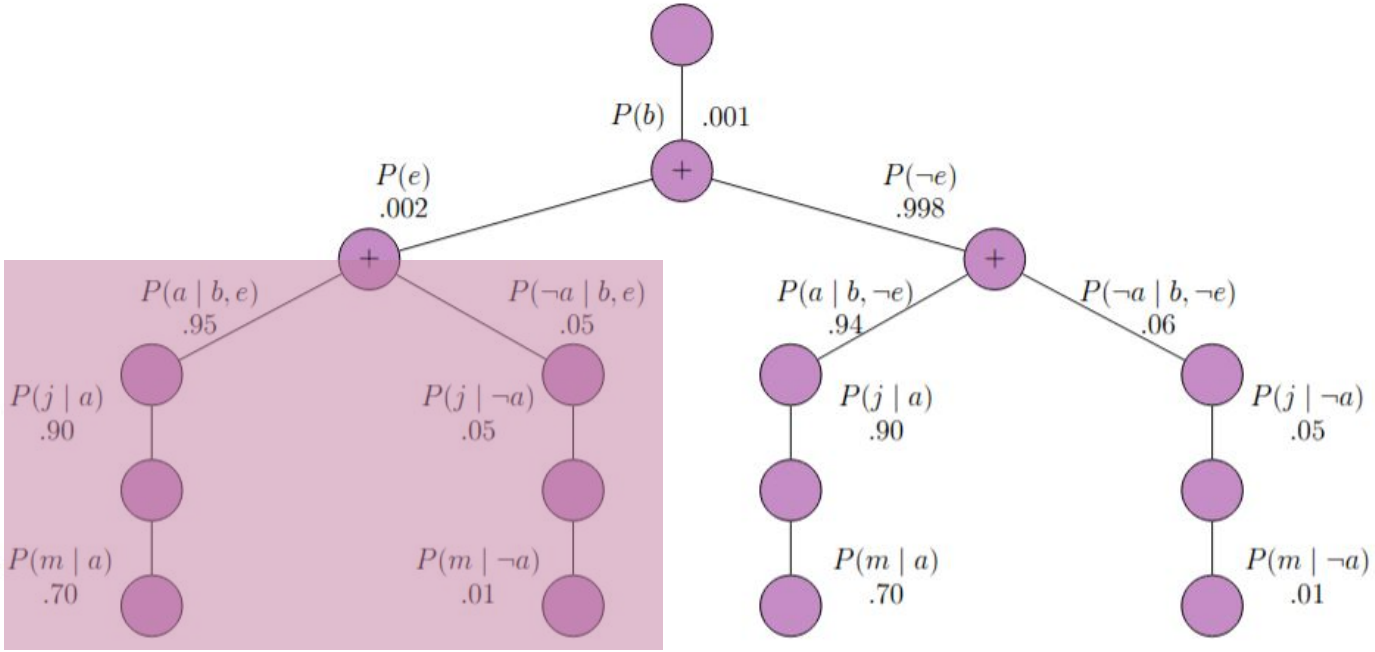
$$\sum_{hidden} P(\ldots)$$

If we have:

- n variables
- k possible values per variable

Complexity: **O (k ^{n})**

Downside of IE Algorithm



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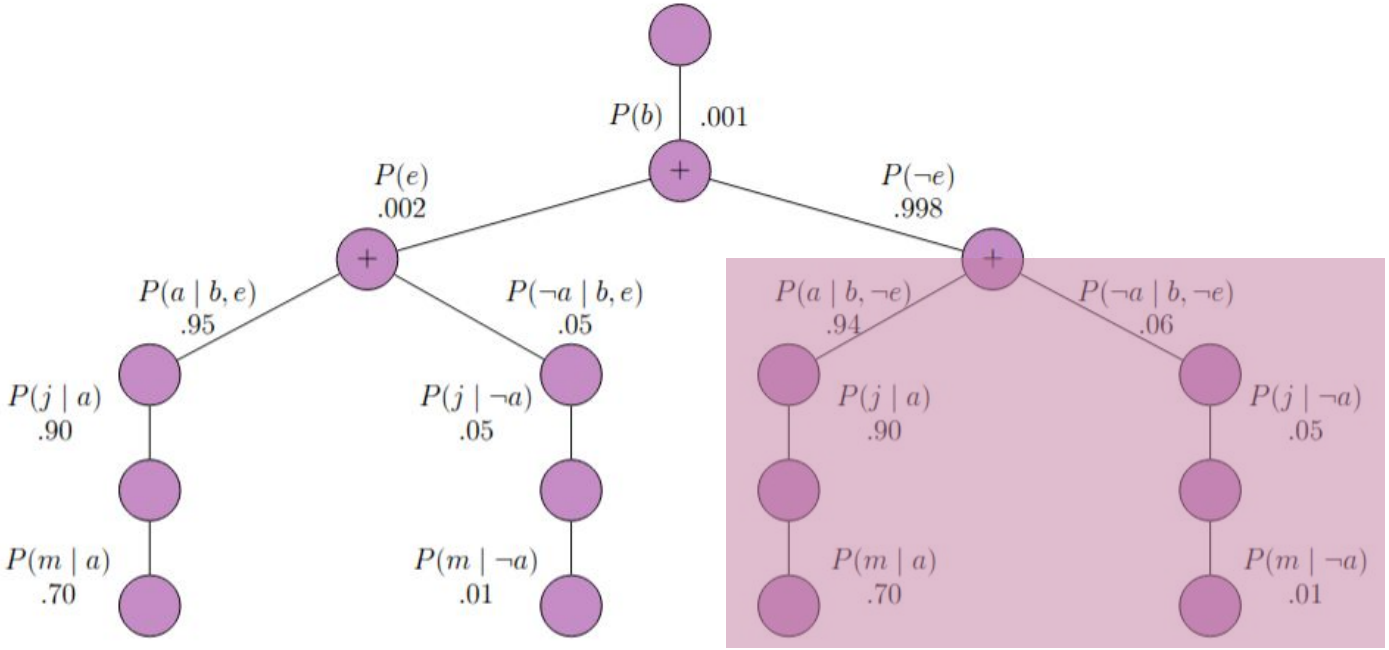

$$P(b \mid j,m) = \alpha P(b) \sum_e P(e) \sum_a P(a \mid b,e) P(j \mid a) P(m \mid a)$$

The Variable Elimination

Improving version of Enumeration algorithm by eliminating repeated calculations

$$\mathbf{P}(B \mid j,m) = \alpha\, \mathbf{P}(B) \sum_{e} P(e) \sum_{a} \mathbf{P}(a \mid B,e)\, P(j \mid a)\, P(m \mid a)\ .$$

Idea - convert probabilities into **factors**, multiply them, and eliminate hidden variables step by step.

The Variable Elimination

Improving version of Enumeration algorithm by eliminating repeated calculations

$$\mathbf{P}(B \mid j,m) = \alpha \underbrace{\mathbf{P}(B)}_{\mathbf{f}_1(B)} \sum_e P(e) \sum_a \mathbf{P}(a \mid B,e) P(j \mid a) P(m \mid a) \ .$$

The Variable Elimination

Improving version of Enumeration algorithm by eliminating repeated calculations

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The Variable Elimination

Improving version of Enumeration algorithm by eliminating repeated calculations

$$\mathbf{P}(B \mid j,m) = \alpha \underbrace{\mathbf{P}(B)}_{\mathbf{f}_1(B)} \sum_e \underbrace{P(e)}_{\mathbf{f}_2(E)} \sum_a \underbrace{\mathbf{P}(a \mid B,e)}_{\mathbf{f}_3(A,B,E)} \underbrace{P(j \mid a)}_{\mathbf{f}_4(A)} P(m \mid a) \ .$$

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$$\mathbf{P}(B \mid j,m) = \alpha \mathbf{f}_1(B) \times \sum_e \mathbf{f}_2(E) \times \sum_a \mathbf{f}_3(A,B,E) \times \mathbf{f}_4(A) \times \mathbf{f}_5(A)$$

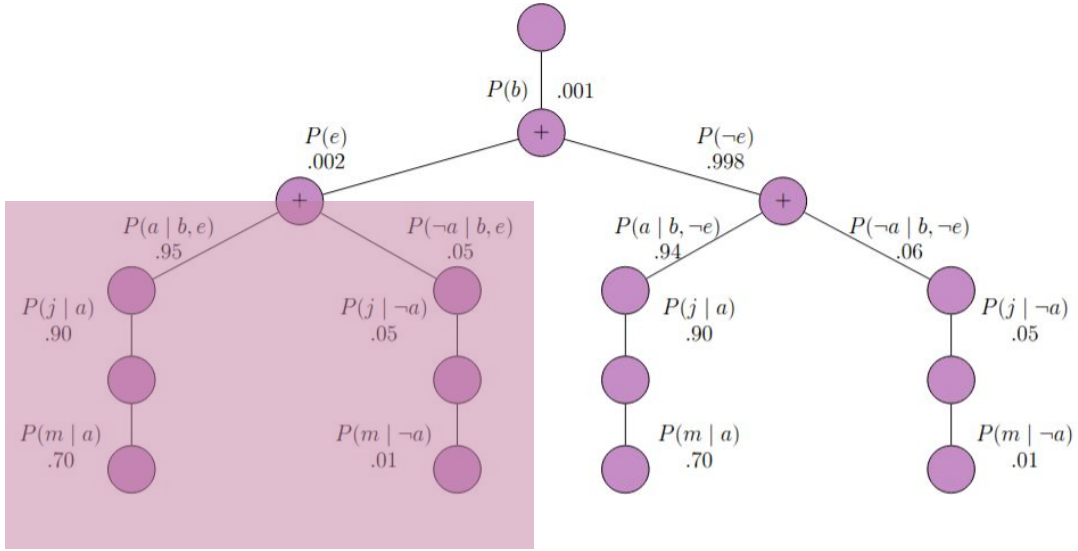
Factors

A factor is a matrix indexed by the values of its argument variables

$$\mathbf{P}(B \mid j,m) = \alpha \underbrace{\mathbf{P}(B)}_{\mathbf{f}_1(B)} \sum_e \underbrace{P(e)}_{\mathbf{f}_2(E)} \sum_a \underbrace{\mathbf{P}(a \mid B,e)}_{\mathbf{f}_3(A,B,E)} \underbrace{P(j \mid a)}_{\mathbf{f}_4(A)} \underbrace{P(m \mid a)}_{\mathbf{f}_5(A)} .$$

$$\mathbf{f}_4(A) = \begin{pmatrix} P(j \mid a) \\ P(j \mid \neg a) \end{pmatrix} = \begin{pmatrix} 0.90 \\ 0.05 \end{pmatrix}$$

Variable Elimination Algorithm Visualization



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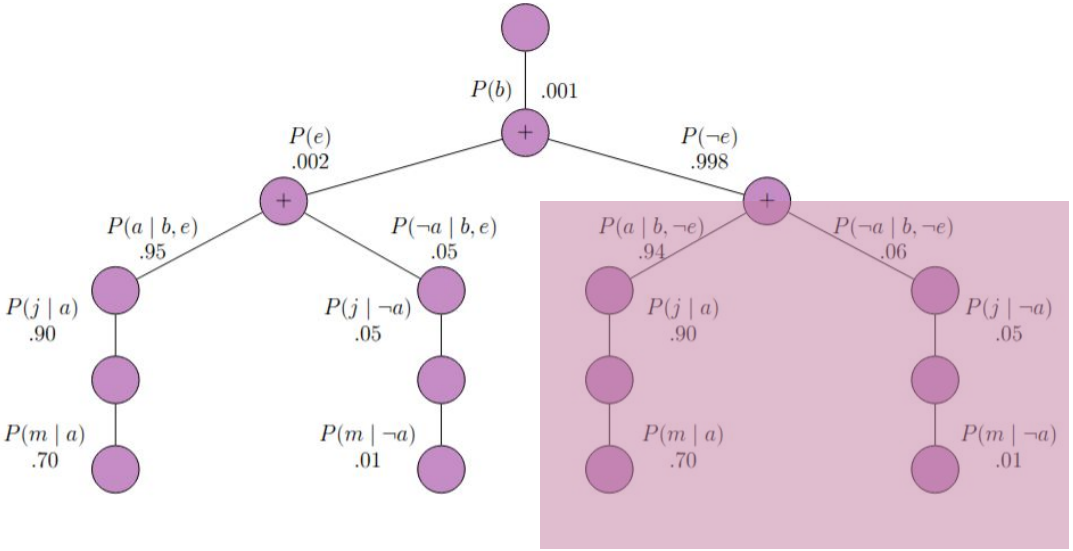

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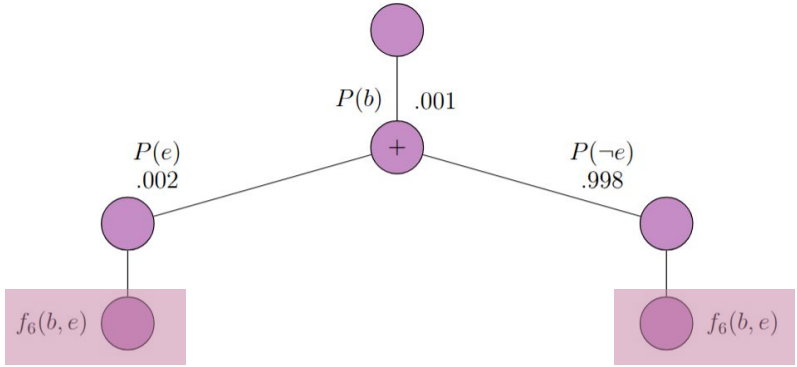

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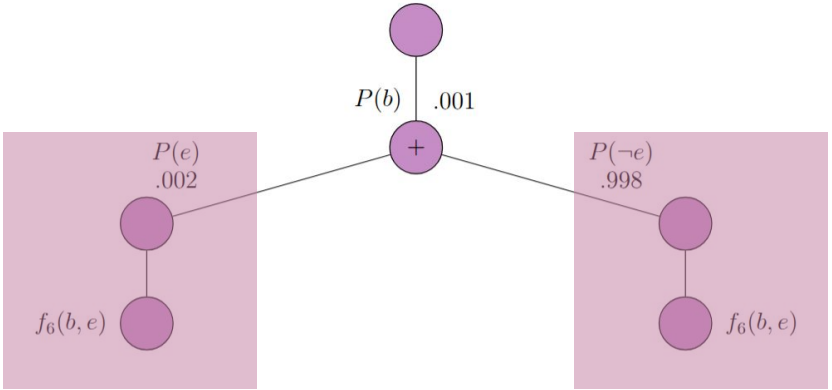

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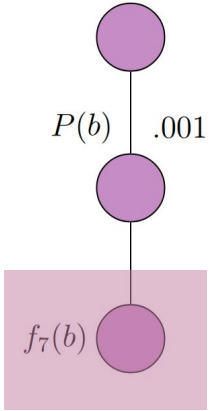
Variable Elimination Algorithm Visualization



$$
\begin{aligned}
\mathbf{P}(B \mid j, m) &= \alpha \mathbf{f}_1(B) \times \sum_e \mathbf{f}_2(E) \times \sum_a \mathbf{f}_3(A, B, E) \times \mathbf{f}_4(A) \times \mathbf{f}_5(A) \\
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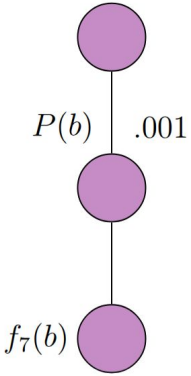
Variable Elimination Algorithm Visualization



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&= \alpha \mathbf{f}_1(B) \times \mathbf{f}_7(B)
\end{aligned}
$$

Complexity of Variable Elimination

$$P(b \mid j,m) = \alpha P(b) \sum_e P(e) \sum_a P(a \mid b,e) P(j \mid a) P(m \mid a)$$

If we have:

- k possible values per variable (in our case 2)
- w = treewidth (maximum number of variables appearing together in a factor)

In dense networks, complexity: $O(k^w)$

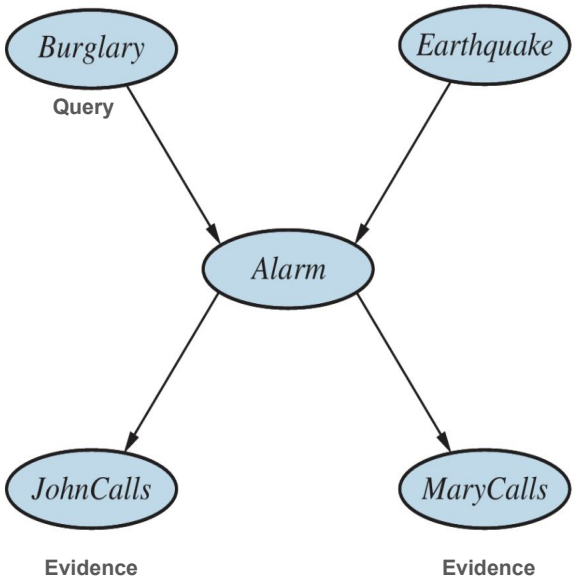
Approximate Inference for Bayesian Networks

What is the probability of a burglary given that John and Mary called?

$$P(B|J=true,M=true)$$

Surprisingly, this probability is not as high as we might expect.

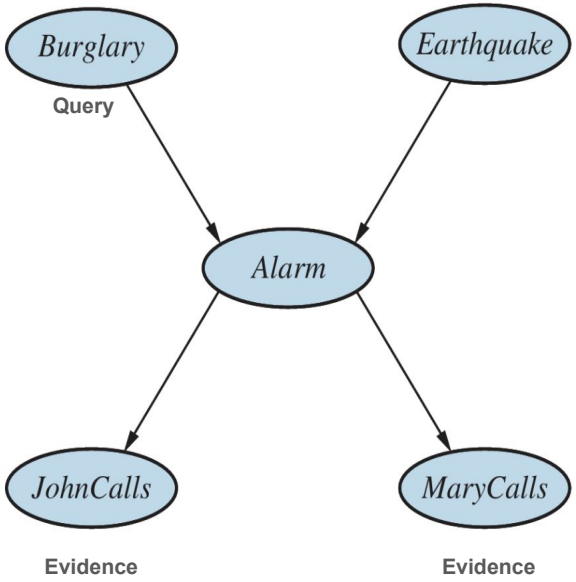
- Query: Burglary
- Evidence: **JohnCalls = true, MaryCalls = true**



How do we compute P(B | J=true, M=true)?

P(B|J=true,M=true)

We cannot compute this directly easily.



Inference by Enumeration

$$\mathbf{P}(B \mid j,m) = \alpha\, \mathbf{P}(B,j,m) = \alpha \sum_e \sum_a \mathbf{P}(B,j,m,e,a)$$

$$\downarrow$$

$$P(b \mid j,m) = \alpha \sum_e \sum_a P(b)P(e)P(a \mid b,e)P(j \mid a)P(m \mid a)$$

$$\downarrow$$

$$P(b \mid j,m) = \alpha P(b) \sum_e P(e) \sum_a P(a \mid b,e)P(j \mid a)P(m \mid a)$$

Why Approximate Inference?

- Exact inference = sum over all hidden variables
- Number of combinations grows exponentially
- Becomes infeasible for large networks

Instead, we use **sampling**.

What is a sample?

A sample = one complete assignment of all variables

It represents one possible world

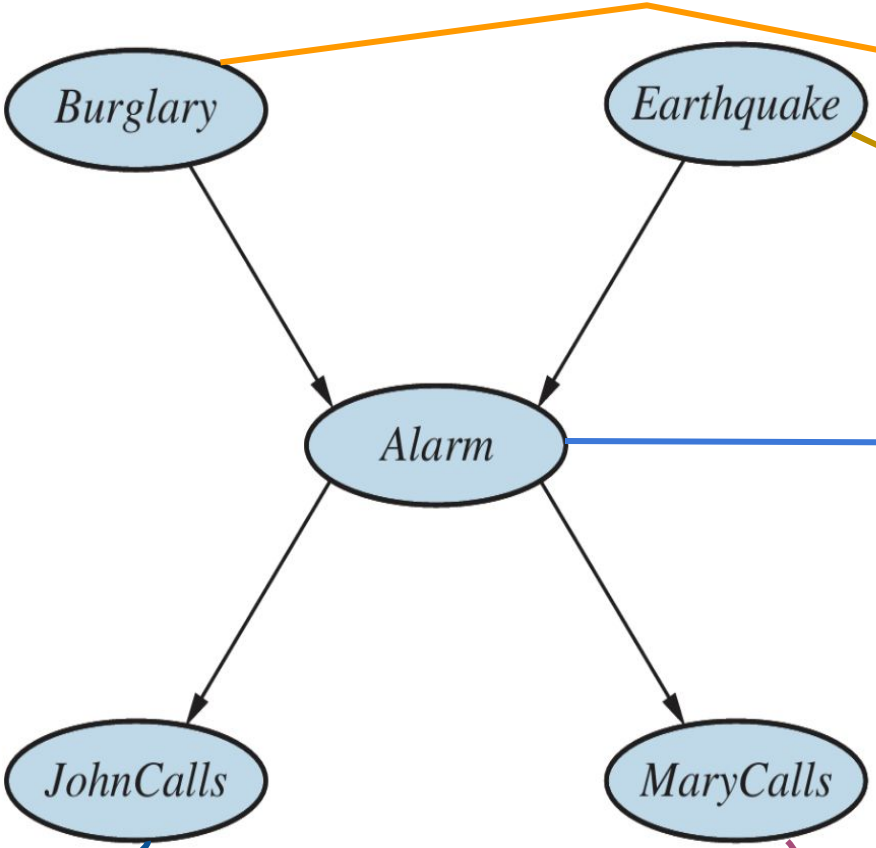
How to make a sample:

1. Generate many samples from the network
2. Keep only samples where:

   J = true and M = true

3. Among those, count how often B = true

[B , E , A , J , M] = [T , F , T , T , T]



P(B) = True= 0.001 | T

P(E) = True= 0.002 | F

B	E	P(A=true|B, E)
T	T	0.95
T	F	0.94
F	T	0.29
F	F	0.001

A	P(J=true|A)
T	0.9
F	0.05

A	P(M=true|A)
T	0.7
F	0.01

[T, F, T, T, T]

Why does sampling give the correct probability?

$P(B|J=true,M=true)$

We approximate probabilities using frequencies:

Probability ≈ Frequency in samples

As number of samples increases:
estimate → true probability

one sample is not enough, many samples are needed

Rejection Sampling

Idea:

- Generate samples from the network
- Keep only samples that match the evidence
- Reject all others

For our example:

Evidence: J=true, M=true

Keep only if: J = true AND M = true

[B , E , A , J , M]

[T , F , T , F , T] **reject**

[T , F , T , T , T] **keep**

[T , F , T , T , F] **reject**

[T , F , T , F , F] **reject**

[F , T , T , T , T] **keep**

Why is rejection sampling inefficient?

- If evidence is rare → most samples are rejected
- Very few useful samples remain
- Waste of computation

[B , E , A , J , M]

[T , F , T , F , T] reject J = F

[T , F , T , T , T] keep

[T , F , T , T , F] reject M = F

[T , F , T , F , F] reject J = F, M = F

[F , T , T , T , T] keep

Likelihood Weighting

How well this sample agrees with the evidence

For our example:

Evidence: J=true, M=true

[B , E , A , J , M]

[T , F , T , T , T] **weight** **HIGH**

[T , T , T , T , T] **weight** **MEDIUM**

[T , F , F , T , T] **weight** **LOW**

Idea:

- Do **not reject samples**
- Fix evidence variables:
  J=true, M=true
- Assign a **weight** to each sample

- Rejection sampling → discard samples
- Likelihood weighting → keep all samples, but weight them

Gibbs Sampling

- *Avoids rejection and weighting issues = We want a method that uses ALL samples AND keeps them useful*

Limitation of previous methods:

- Rejection sampling → wastes many samples
- Likelihood weighting → weights can become very small

Idea:

- Start with a full assignment
- Keep evidence fixed
- Update **one variable at a time**

Gibbs Sampling

Gibbs Sampling is based on a Markov Chain.

This means that each new sample depends only on the previous one, not on the full history.

Instead of generating new samples from scratch

1. take one sample

2. update it step by step

Evidence: J = true, M = true

“Let’s slightly change it to make it more realistic”

We start somewhere (random) because **Gibbs sampling is iterative.**

[B , E , A , J , M]

[T , F , T , T , T]

Gibbs Sampling

Evidence: J = true, M = true

Given everything else, what should A be?

[B , E , A , J , M]

[T , F , T , T , T]

Pick ONE variable (e.g., A)

P (A | B , E , J , M)

[T , F , F , T , T]

We resample A based on its probability given the current values of the other variables - not to force a change, but to reflect uncertainty.

Gibbs Sampling

Evidence: J = true, M = true

[B , E , A , J , M]

[T , F , T , T , T]

Pick ONE variable (e.g., A)

P (A | B , E , J , M)

Given everything else, what should A be?

We resample A based on its probability given the current values of the other variables - not to force a change, but to reflect uncertainty.

Update the state to:

[T , F , F , T , T]

Gibbs Sampling

Evidence: J = true, M = true

[B , E , A , J , M]

[T , F , T , T , T]

Pick ONE variable (e.g., A)

P (A | B , E , J , M)

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We resample A based on its probability given the current values of the other variables - not to force a change, but to reflect uncertainty.

Update the state to:

[T , F , F , T , T]

Given everything else, what should B be?

Pick ONE variable (e.g., B)

P (B | E , A, J , M)

Gibbs Sampling

Evidence: J = true, M = true

[B , E , A , J , M]

[T , F , T , T , T]

Given everything else, what should A be?

Pick ONE variable (e.g., A)

P (A | B , E , J , M)

We resample A based on its probability given the current values of the other variables - not to force a change, but to reflect uncertainty.

Update the state to:

[T , F , F , T , T]

Given everything else, what should B be?

Pick ONE variable (e.g., B)

P (B | E , A, J , M)

Update B → Update E

then A again
then B again…

This repeats many times → After many updates, the samples approximate the true distribution

Thank you for your attention!

Questions & Discussion

Question 1 - Bayesian Networks

We can already store all probabilities in one big table. So why do we still need Bayesian Networks?

Question 2 - Bayesian Networks

Consider the variables: **Disease, Symptoms, TestResult**

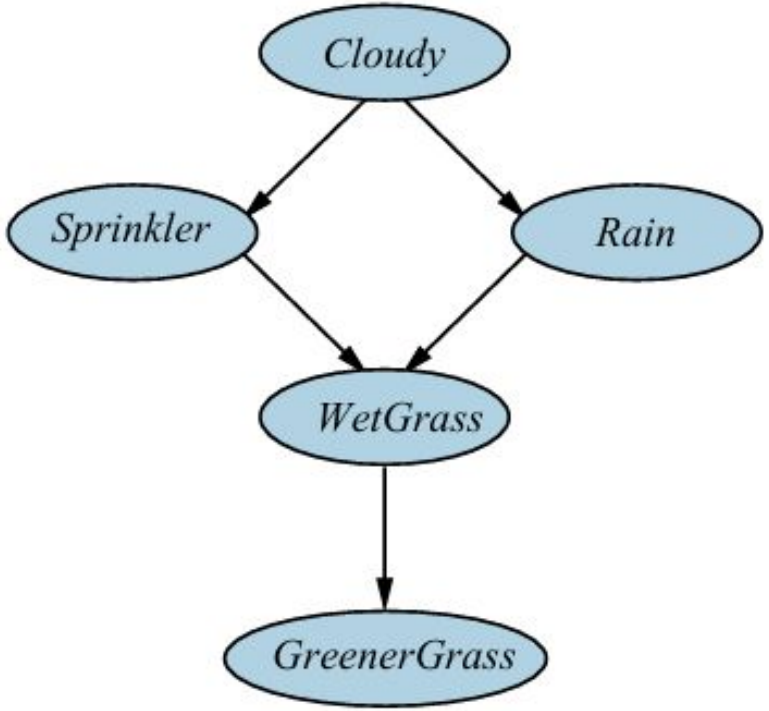
- In what order would you construct the Bayesian Network using causal order?
- Define the relationships between these variables.

Question 3 - Bayesian Networks

If we don’t choose an efficient order for variables when building the network, does the probability distribution become wrong?

Question 4 - Bayesian Networks

Consider the following network. What is the Markov Blanket of variable WetGrass?



Question 5 - Exact Inference

What was the goal of Exact inference?

Question 6 - Inference by enumeration

In our example

```
P ( Burglary | JohnCalls , MaryCalls )
```

what type of variables can you detect?

Question 7 - Inference by enumeration

Were there any other variables we needed to consider?

Question 8 - Inference by enumeration

What if we ignore hidden variables, can we still compute the probability?

Question 9 - Variable Elimination

What exactly makes Variable Elimination faster?

Question 10 - Exact vs Approximate

Why don’t we always use exact inference if it gives the correct answer?

Question 11 - Gibbs Core Idea

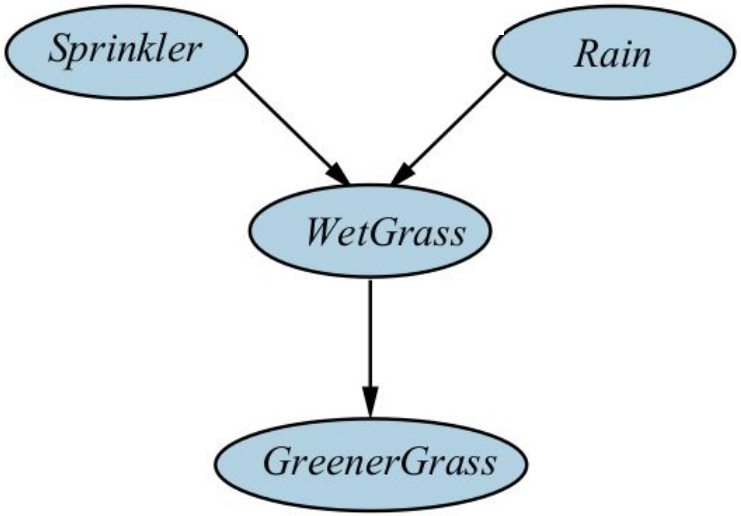
What is the main idea behind Gibbs sampling?

Question 12 - Rejection Sampling

Why rejection sampling is inefficient?

Question 13 - Sampling

Do you think sampling methods are reliable enough for critical systems?

Extra example

What's the probability that Sprinkler was on if we detect that Grass is greener?